

AI-Assisted Medical Diagnosis:

Deep Learning-Based AI Radiology Systems in Healthcare

A Report on the Application of Artificial Intelligence in the Healthcare Domain

COURSE : ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS

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NAME : YAMEENULLAH M H

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Executive Summary

Artificial Intelligence is transforming how diseases are detected and diagnosed. This report examines AI-based medical diagnosis, with a specific focus on AI radiology systems that use deep learning to analyse X-rays, CT scans, and MRI scans for signs of disease such as pneumonia, tuberculosis, diabetic retinopathy, stroke, and cancer.

The report explains the healthcare problems these systems address — including specialist shortages, diagnostic delay, and inconsistency in manual review — and why AI, rather than traditional software or purely manual processes, is well suited to solving them. It describes the core AI technologies involved (Convolutional Neural Networks, transfer learning, computer vision, and explainable AI), and walks through the working mechanism from image acquisition to a clinician-reviewed diagnostic output.

The report also surveys real-world deployments across hospitals and screening programmes globally, quantifies the impact and benefits observed, and critically examines the challenges that remain — particularly around data bias, explainability, and regulatory accountability. It concludes that AI is most effective and most responsibly used as a decision-support tool that augments, rather than replaces, human clinical judgement, and outlines the emerging trends likely to shape the next generation of AI-assisted diagnostic tools.

1. Title of AI Application

Title: “AI-Based Medical Diagnosis using Deep Learning – AI Radiology Systems for Automated Disease Detection”.

AI Domain: Healthcare.

This report examines a leading contemporary AI application in the healthcare sector — AI-powered radiology systems that use deep learning models to analyse medical images such as X-rays, CT scans, and MRI scans in order to detect diseases faster and more accurately than traditional manual review alone. Systems of this kind, such as those developed for detecting pneumonia, tuberculosis, diabetic retinopathy, and various forms of cancer, represent one of the most mature and widely deployed uses of artificial intelligence in modern clinical practice.

2. Domain Selection

The selected domain for this study is Healthcare, one of the six domains identified for scenario-based research (the others being Finance, Retail, Autonomous Vehicles, Robotics, and Manufacturing). Healthcare was chosen because it is a domain where AI adoption has an immediate and measurable impact on human wellbeing, and because AI-assisted diagnosis exemplifies the broader trend of AI systems augmenting — rather than replacing — expert human judgement.

Within healthcare, AI applications span diagnosis, treatment planning, patient monitoring, drug discovery, and virtual health assistance. This report focuses specifically on the diagnostic sub-domain, and in particular on AI radiology and medical imaging, because it demonstrates a complete pipeline from raw sensor data (medical images) to a clinically actionable output (a diagnostic recommendation).

3. Introduction and Background

Healthcare organisations worldwide are increasingly adopting Artificial Intelligence to improve diagnosis, treatment, and patient monitoring. Medical imaging — X-rays, computed tomography (CT), magnetic resonance imaging (MRI), and ultrasound — has traditionally depended on radiologists manually interpreting scans, a process that is accurate but time-consuming and subject to human fatigue and variability.

AI-based radiology systems use deep learning, particularly Convolutional Neural Networks (CNNs), trained on large datasets of labelled medical images to recognise patterns associated with specific conditions. Early systems in this space, such as Google DeepMind’s work on diabetic retinopathy screening and IBM Watson Health’s oncology tools, demonstrated that trained models could match or exceed specialist-level accuracy on narrowly defined diagnostic tasks. Since then, regulatory-approved AI diagnostic tools (for example, those cleared by the U.S. FDA for stroke detection, breast cancer screening, and diabetic eye disease) have moved from research prototypes into routine clinical workflows in hospitals and diagnostic centres globally.

The purpose of this AI application is to act as a “second reader” or first-pass triage tool: flagging scans that show signs of disease, prioritising urgent cases, and reducing the diagnostic workload on human radiologists, while the final clinical decision remains with a qualified physician.

3.1 Historical Evolution

The use of computers to assist medical image interpretation is not new — Computer-Aided Detection (CAD) systems using rule-based image processing were introduced in mammography as early as the 1990s. However, these early systems relied on hand-crafted image features and had limited accuracy, generating a high rate of false positives that made clinicians reluctant to depend on them.

The shift towards modern deep-learning-based diagnostic AI began around 2012, when breakthroughs in CNN architectures (originating from general-purpose computer vision research) demonstrated dramatic improvements in image classification accuracy. Researchers quickly adapted these architectures to medical imaging, and by the mid-2010s, peer-reviewed studies were showing that trained CNNs could match specialist-level performance on specific, well-defined diagnostic tasks such as skin cancer classification and diabetic retinopathy grading. This period marked the transition of AI in healthcare from an experimental research topic to a commercially viable clinical technology.

3.2 Scope of This Report

This report focuses on diagnostic imaging AI as a representative case study of the broader trend of AI adoption in healthcare. While AI is also being applied to drug discovery, hospital operations, virtual nursing assistants, and remote patient monitoring, medical imaging was selected because it offers the clearest illustration of a complete AI pipeline — from raw data collection, through model training and inference, to a concrete clinical decision — making it well suited to an in-depth technical and ethical analysis.

4. Problem Identification

The real-world problem addressed by AI-based medical diagnosis systems can be broken down into several concrete challenges faced by healthcare systems today:

- **Shortage of specialists:** Many regions, particularly rural and low-income areas, have very few qualified radiologists relative to the population, leading to long waiting times for scan interpretation.
- **Diagnostic delay:** Manual review of every scan is time-intensive, and delays in detecting conditions such as cancer or stroke directly reduce patient survival and recovery outcomes.
- **Human error and fatigue:** Radiologists reviewing hundreds of images a day are susceptible to fatigue-related oversight, and subtle early-stage abnormalities can be missed.
- **Inconsistent diagnostic quality:** Interpretation can vary between practitioners of different experience levels, leading to inconsistent diagnostic accuracy across facilities.
- **Rising imaging volumes:** The number of scans ordered per patient has grown steadily, increasing pressure on radiology departments without a proportional increase in specialist staff.

These problems collectively point to a need for a scalable, consistent, and fast diagnostic support tool — the gap that AI-based radiology systems are designed to fill.

4.1 Illustrative Scenario

Consider a district hospital that receives 300 chest X-rays a day but has only two radiologists on staff. Even working continuously, the radiology team can only thoroughly review a fraction of these scans within a clinically useful timeframe, meaning some patients with early signs of pneumonia or tuberculosis may wait days for a formal diagnosis. In such a scenario, an AI triage tool that automatically flags the scans most likely to show abnormalities allows the limited specialist staff to prioritise the most urgent cases first, directly reducing the risk of harm caused by diagnostic delay.

5. Need for AI Solution

Traditional diagnostic approaches rely entirely on human expertise, which, while highly skilled, cannot scale to match growing imaging volumes without a proportional increase in trained personnel — a resource that is expensive and slow to develop. AI is required in this context for several reasons:

- **Speed:** A trained deep learning model can screen a scan in seconds, compared to minutes for manual review, enabling rapid triage of urgent cases.
- **Scalability:** Once trained, an AI model can be deployed across many facilities simultaneously without requiring additional specialist hiring.
- **Consistency:** AI models apply the same learned criteria to every image, reducing variability introduced by fatigue or differing experience levels.
- **Pattern recognition at scale:** Deep learning models can detect subtle statistical patterns across thousands of pixels that may not be immediately obvious to the human eye, particularly in early-stage disease.
- **Cost-efficiency:** Automating first-pass screening can reduce the overall cost of diagnostic workflows, especially in high-volume screening programmes such as population-wide diabetic retinopathy screening.

Importantly, AI improves upon — rather than replaces — traditional approaches: it acts as a decision-support layer that augments radiologists' judgement, catching cases that might otherwise be deprioritised and allowing specialists to focus their attention on the most complex or ambiguous scans.

5.1 Why Not a Simpler, Rule-Based System?

It is worth noting why AI — specifically deep learning — is needed rather than a simpler, rule-based software system. Medical images contain highly complex, subtle visual patterns that do not reduce well to a small set of hand-written rules; the same disease can present differently across patients, imaging equipment, and disease stages. Deep learning models are able to learn these complex, non-linear patterns directly from large volumes of labelled examples, achieving a level of pattern recognition that rule-based systems — which rely on explicitly programmed logic — have consistently failed to match in image-heavy diagnostic tasks.

6. AI Technologies Used

AI-based radiology systems draw on several core AI and machine learning techniques, used together as part of an end-to-end diagnostic pipeline:

- **Deep Learning:** The foundational technique, using multi-layered neural networks capable of learning complex representations directly from raw image data.
- **Convolutional Neural Networks (CNNs):** The primary architecture used for image analysis, applying learned filters to detect edges, textures, and shapes that correspond to anatomical structures and abnormalities. Architectures such as ResNet, DenseNet, and U-Net are commonly used.
- **Computer Vision:** Techniques for image pre-processing, segmentation (isolating organs or regions of interest), and classification of abnormal findings.
- **Transfer Learning:** Pre-trained models (trained on large general-purpose image datasets) are fine-tuned on smaller, specialised medical imaging datasets, which is essential given the relative scarcity of large labelled medical datasets.
- **Natural Language Processing (NLP):** Used in some systems to automatically generate structured diagnostic reports from image findings, and to extract information from unstructured clinical notes.
- **Explainable AI (XAI):** Techniques such as saliency maps and Grad-CAM are used to visually highlight the regions of an image that most influenced the model's prediction, which is critical for clinician trust and regulatory approval.

6.1 Summary of Techniques and Their Roles

Technique	Category	Primary Role in the Pipeline
CNN (ResNet, U-Net)	Deep Learning	Extracts visual features and classifies or segments abnormalities
Transfer Learning	Model Training	Adapts general pre-trained models to limited medical datasets
Computer Vision	Image Processing	Handles segmentation, normalisation, and region-of-interest detection
NLP	Language AI	Auto-generates structured diagnostic reports from findings
Grad-CAM / XAI	Explainability	Highlights the image regions driving the model's prediction

7. Working Mechanism

An AI radiology system typically operates through the following stages, from raw scan capture to a clinically usable output:

- **Image Acquisition:** A scan (X-ray, CT, MRI, etc.) is captured using standard hospital imaging equipment and stored in DICOM (Digital Imaging and Communications in Medicine) format.
- **Pre-processing:** The image is normalised, resized, and cleaned (removing noise, adjusting contrast) to a standard format suitable for the model.
- **Feature Extraction:** The CNN passes the image through successive convolutional layers, each learning increasingly abstract features — from simple edges in early layers to complex shapes indicative of specific pathologies in deeper layers.
- **Classification / Segmentation:** The model outputs either a classification (e.g., “pneumonia present” / “no abnormality detected”) with an associated confidence score, or a segmentation map highlighting the precise location of an abnormality.
- **Decision Support Output:** The result is presented to a radiologist alongside the original scan, often with a heatmap overlay indicating the region of concern, functioning as a second opinion rather than an automatic diagnosis.
- **Clinician Review and Final Decision:** A qualified radiologist or physician reviews the AI’s output together with the original image and other clinical information before confirming a diagnosis and treatment plan.

This human-in-the-loop design is a deliberate and important feature: the AI system accelerates and supports the diagnostic process, but accountability for the final clinical decision remains with the treating physician.

7.1 Worked Example: Chest X-ray Triage

To illustrate the working mechanism concretely, consider a chest X-ray triage system. When a new X-ray is captured, it is automatically sent to the AI model within seconds of acquisition. The model processes the image and returns a probability score for several conditions (for example, a 92% likelihood of pneumonia). If this score exceeds a clinically validated threshold, the case is automatically moved to the top of the radiologist’s workload and flagged with a heatmap showing the suspected affected region of the lung. The radiologist then opens the case, reviews both the original image and the AI-highlighted region, and confirms or overrides the AI’s suggestion before finalising the report. The entire AI processing step typically completes in under 30 seconds, compared to an average wait of several hours for a case to reach the top of a manual workload.

8. Data Sources and Processing

AI radiology models are trained on large datasets of medical images that have been labelled by expert radiologists to indicate the presence, absence, and location of specific conditions. Key data sources and processing steps include:

- **Public and institutional datasets:** Datasets such as ChestX-ray14, MIMIC-CXR, and the Radiological Society of North America (RSNA) datasets provide large volumes of labelled scans used for research and model development.
- **Hospital PACS systems:** In deployed clinical settings, models are integrated with a hospital’s Picture Archiving and Communication System (PACS), from which new scans are automatically routed to the AI system for analysis.

- **Data labelling:** Radiologists annotate training images with disease labels and, where relevant, bounding boxes or pixel-level masks identifying the affected region, forming the “ground truth” used to train the model.
- **Data augmentation:** Techniques such as rotation, flipping, and contrast adjustment are applied to training images to artificially increase dataset diversity and reduce overfitting, given the relatively limited size of labelled medical datasets.
- **Data privacy and de-identification:** Patient data is anonymised in compliance with regulations such as HIPAA (United States) and equivalent data protection laws, removing personally identifiable information before it is used for model training.

The quality, diversity, and representativeness of this training data directly determines the reliability of the resulting model — a dataset skewed towards a particular demographic or imaging equipment type can lead to reduced accuracy for underrepresented patient groups.

8.1 Representative Public Datasets

Dataset	Imaging Type	Primary Use
ChestX-ray14	Chest X-ray	Training models to detect 14 thoracic conditions including pneumonia
MIMIC-CXR	Chest X-ray + reports	Combining images with free-text radiology reports for NLP research
RSNA Datasets	X-ray, CT, MRI	Benchmarking model performance in open research challenges
EyePACS	Retinal fundus images	Training and validating diabetic retinopathy screening models

These datasets are typically supplemented with institution-specific data once a model is being fine-tuned for deployment in a particular hospital system, since imaging equipment, patient demographics, and clinical protocols vary meaningfully across regions and providers.

Comparison: AI-Assisted vs Traditional Diagnosis

To place the preceding technical discussion in context, the table below compares AI-assisted diagnostic imaging with the traditional, fully manual approach across several practical dimensions relevant to hospitals and screening programmes.

Dimension	Traditional Manual Diagnosis	AI-Assisted Diagnosis
Speed	Minutes to hours per scan, dependent on workload	Seconds per scan for first-pass screening
Consistency	Varies with experience, fatigue, and workload	Consistent criteria applied to every scan
Scalability	Limited by number of trained specialists	Scales across facilities with minimal added staff
Transparency	Reasoning is fully explainable by the clinician	Reasoning is partially opaque without XAI tools
Accountability	Clearly rests with the diagnosing clinician	Shared between clinician, developer, and institution

This comparison underscores why leading healthcare institutions treat AI as a complement to, rather than a substitute for, clinical expertise: each approach has distinct strengths, and the combination — AI for speed and consistency, humans for contextual judgement and accountability — currently produces better outcomes than either approach alone.

9. Real-World Implementation

AI-based diagnostic imaging tools have moved from research settings into active clinical use across multiple healthcare systems:

- **Diabetic retinopathy screening:** AI systems are used in screening programmes to automatically grade retinal images for signs of diabetic eye disease, particularly valuable in regions with limited access to ophthalmologists.
- **Chest X-ray triage:** Hospitals use AI tools to flag chest X-rays showing signs of pneumonia, tuberculosis, or lung nodules, prioritising urgent cases in radiologists' worklists.
- **Stroke detection:** AI software analyses CT and MRI brain scans to rapidly identify signs of stroke, helping clinical teams make faster treatment decisions in time-critical situations.
- **Cancer screening:** AI-assisted mammography analysis is used in breast cancer screening programmes to help detect early-stage tumours that may be difficult to spot on visual inspection alone.
- **COVID-19 and respiratory disease detection:** During the COVID-19 pandemic, several AI imaging tools were deployed to assist in rapidly assessing lung CT scans for signs of viral pneumonia.

These systems are typically deployed as an integrated software layer within existing hospital IT infrastructure, working alongside — not replacing — radiology departments, and are subject to regulatory clearance (such as FDA or CE marking) before clinical use.

9.1 Selected Deployment Examples

Application Area	Condition Targeted	Typical Deployment Setting
Retinal screening	Diabetic retinopathy	Primary care clinics and community screening camps
Chest imaging	Pneumonia, tuberculosis	Hospital radiology departments, emergency triage
Neuroimaging	Acute stroke	Emergency stroke units for rapid treatment decisions
Mammography	Breast cancer	National and regional breast cancer screening programmes
Lung CT analysis	Viral pneumonia (e.g., COVID-19)	Hospitals during high-volume respiratory outbreaks

Case Study: Diabetic Retinopathy Screening at Scale

One of the clearest illustrations of AI-assisted diagnosis in practice is automated diabetic retinopathy (DR) screening. Diabetic retinopathy is a leading cause of preventable blindness worldwide, and early detection through regular retinal screening can prevent vision loss in the large majority of cases. However, many regions with a high burden of diabetes have very few ophthalmologists available to perform manual retinal screening at the scale required.

To address this gap, several healthcare systems have deployed deep learning models trained on large sets of labelled retinal fundus photographs to automatically grade images for signs of DR. In a typical deployment, a technician (who does not need to be a specialist) captures a retinal photograph using a handheld or desktop fundus camera at a primary care clinic. The image is uploaded to an AI system, which analyses it within seconds and returns a grading indicating whether the patient shows no DR, mild-to-moderate DR requiring routine follow-up, or referable DR requiring urgent specialist review.

This workflow has enabled screening programmes to reach patients in areas with no local ophthalmologist, using existing primary care infrastructure and lower-skilled technicians for image capture, while reserving scarce specialist time for the smaller number of patients whose scans are flagged as requiring urgent attention. Published studies evaluating such systems have reported sensitivity and specificity for detecting referable disease that is comparable to trained human graders, supporting their use as a validated first-line screening tool within a broader clinical pathway that still involves ophthalmologist confirmation for flagged cases.

This case illustrates several themes discussed throughout this report in a single, concrete example: the problem (specialist shortage), the AI solution (automated grading from a CNN-based model), the working mechanism (image capture, AI inference, referral decision), and the real-world impact (expanded screening access and earlier detection) all come together in a deployed system that is already improving patient outcomes at population scale.

10. Impact and Benefits

The adoption of AI in medical diagnosis has produced measurable improvements across several dimensions of healthcare delivery:

- **Efficiency:** Automated first-pass screening significantly reduces the time required to review large volumes of scans, allowing radiologists to focus on complex or ambiguous cases.
- **Accuracy:** In several well-studied applications, AI models have demonstrated diagnostic accuracy comparable to, or in narrow tasks exceeding, that of individual specialists, particularly in detecting subtle early-stage abnormalities.
- **Cost reduction:** Faster, more efficient screening reduces the overall cost per diagnosis, particularly valuable in large-scale public health screening programmes.
- **Improved access:** AI tools can extend diagnostic capability to regions with a shortage of specialist radiologists, improving healthcare equity.

- **Earlier detection:** Faster, more consistent screening increases the likelihood of catching disease at an earlier, more treatable stage, directly improving patient outcomes.
- **Reduced clinician burnout:** By automating repetitive first-pass review tasks, AI tools can reduce the workload-related fatigue experienced by radiology staff.

10.1 Broader Health-System Impact

Beyond individual patient encounters, the cumulative effect of faster and more consistent screening is felt at the health-system level. Screening programmes that were previously limited by specialist availability can be scaled to cover larger populations, catching disease earlier across a wider group of patients rather than only those who could access a specialist quickly. This shift from reactive to proactive, population-scale screening is one of the most significant systemic benefits associated with AI-assisted diagnosis, and is particularly valuable for chronic, slow-progressing conditions such as diabetic eye disease, where early intervention meaningfully changes long-term outcomes.

11. Challenges and Limitations

Despite its benefits, the deployment of AI in medical diagnosis raises significant technical, ethical, and practical challenges:

- **Data bias:** Models trained on non-representative datasets can perform poorly for underrepresented patient groups, raising concerns about healthcare equity.
- **Explainability:** Deep learning models are often described as “black boxes,” making it difficult for clinicians to fully understand why a particular prediction was made, which can affect trust and clinical adoption.
- **Regulatory and legal accountability:** Questions remain about liability when an AI system contributes to a missed or incorrect diagnosis, and regulatory frameworks are still evolving to address this.
- **Data privacy and security:** Handling sensitive patient imaging data requires strict compliance with privacy regulations and robust protection against data breaches.
- **Generalisability:** Models trained on data from one hospital or imaging device may not generalise well to scans from different equipment or patient populations.
- **Over-reliance risk:** There is a risk that clinicians may over-trust AI outputs over time, potentially reducing the critical scrutiny that human oversight is meant to provide.
- **Integration cost:** Deploying AI tools requires investment in compatible IT infrastructure, staff training, and ongoing model validation, which can be a barrier for smaller or resource-constrained healthcare providers.

11.1 Benefits and Challenges at a Glance

Key Benefits	Key Challenges
Faster first-pass screening and triage	Risk of bias from unrepresentative training data

Consistent, fatigue-free image analysis	Limited explainability of deep learning outputs
Extends specialist-level screening to under-served areas	Unclear liability when AI contributes to an error
Earlier detection of disease, improving outcomes	Sensitive patient data requires strict privacy safeguards
Reduced workload and burnout for radiology staff	High cost of infrastructure and staff training to deploy

11.2 Ethical Considerations

Beyond the technical challenges above, AI-based diagnosis raises broader ethical questions that healthcare providers must navigate. Informed consent becomes more complex when a patient's diagnosis is partly influenced by an automated system whose reasoning may not be fully transparent even to the treating clinician. Questions of fairness arise when a model performs less accurately for certain demographic groups due to skewed training data, potentially widening rather than narrowing existing healthcare disparities. Finally, the increasing use of patient imaging data to train commercial AI products raises questions about data ownership and whether patients should have a say in how their anonymised medical data is used to build products that may later be sold back to the healthcare system.

12. Future Scope and Emerging Trends

The field of AI-assisted diagnosis continues to evolve rapidly, with several emerging directions likely to shape its future:

- **Multimodal AI:** Combining imaging data with electronic health records, genomic data, and clinical notes to produce more holistic and personalised diagnostic assessments.
- **Federated learning:** Training models across multiple hospitals without centralising sensitive patient data, improving model generalisability while preserving privacy.
- **Real-time diagnostic support:** Integration of AI directly into imaging hardware for instant analysis at the point of scan capture, reducing turnaround time further.
- **Improved explainability:** Continued research into explainable AI techniques to make model reasoning more transparent and clinically interpretable.
- **Broader disease coverage:** Expansion of AI diagnostic tools beyond currently well-studied conditions into rarer diseases and more complex multi-condition diagnosis.
- **Global health applications:** Wider deployment of low-cost AI screening tools in low-resource settings to address global disparities in access to specialist healthcare.

As regulatory frameworks mature and explainability techniques improve, AI is expected to become an increasingly standard component of diagnostic workflows across a broader range of medical specialities, further shifting the role of clinicians towards oversight, interpretation, and patient communication.

12.1 The Evolving Role of the Clinician

As AI tools take on a larger share of first-pass image review, the role of the radiologist and physician is expected to shift rather than shrink — from primarily manual image interpretation towards oversight of AI outputs, handling complex or ambiguous cases that fall outside a model's trained scope, and communicating findings and treatment decisions to patients. This evolving division of labour between human expertise and automated pattern recognition is likely to define the next phase of AI adoption across healthcare more broadly, not just in diagnostic imaging.

13. Conclusion

AI-based medical diagnosis, particularly in the field of radiology, represents one of the most impactful and mature applications of artificial intelligence in healthcare today. By combining deep learning, computer vision, and large-scale medical imaging datasets, these systems address real and pressing problems: specialist shortages, diagnostic delay, and inconsistency in manual image review.

The technology delivers measurable benefits in speed, consistency, and diagnostic support, while significant challenges remain around bias, explainability, regulatory accountability, and equitable access. The most effective and responsible implementations position AI not as a replacement for clinical judgement, but as a decision-support tool that works alongside radiologists and physicians — improving efficiency and outcomes while keeping human expertise and accountability at the centre of patient care. As the underlying technology and regulatory landscape continue to mature, AI-assisted diagnosis is likely to become an increasingly standard and trusted part of global healthcare delivery.

From a broader research perspective, this case study also demonstrates a pattern that is likely to recur across other domains examined in parallel studies — finance, retail, autonomous vehicles, robotics, and manufacturing: AI succeeds most durably where it is designed to augment scarce human expertise rather than replace it outright, where its outputs remain interpretable and auditable, and where clear accountability structures are maintained alongside the technology's deployment. Healthcare, given the direct human stakes involved, offers perhaps the clearest and most instructive example of this principle in action.

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Annexure A

Scenario-Based Research Assessment

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Criterion	Weightage	Marks Obtained
Identification of Latest AI Application	10%	
Problem Analysis and Domain Understanding	15%	
AI Technologies and Working Mechanism	20%	
Research Quality and Literature Review	10%	
Real-World Implementation and Case Analysis	10%	
Impact, Challenges, and Future Scope	15%	
Originality and Critical Thinking	10%	
Organization, Presentation, and Documentation	10%	
Total	100%	

Signature of the Faculty: _____

Total Marks Awarded: _____